🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.4

Prime Factorization

Lesson 6-13 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write a number as a product of its prime factors using a factor tree.
Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Prime Factorization | **Prime number** | **Composite number** | **Factor** | **Factor tree** | **Exponent**

👆 *Tap any word to see what it means and a picture.*

1. Writing a number as ______________ numbers multiplied together is its prime factorization.

2. A whole number greater than 1 with exactly two factors, 1 and itself, is a ______________ number.

3. A whole number greater than 1 that has more than two factors is a ______________ number.

4. A number that divides evenly into another number is called a ______________.

5 I can break a number into its prime factors step by step using a

________________ .

6 In 2^3, the small number 3 is the ________________ and it shows 2 is multiplied 3 times.

Write About the Math

Starting from its prime factorization, how can the gardener use the factors of 72 to decide which rectangular grids will use all 72 seed pods?

Empezando por su descomposición en factores primos, ¿cómo puede la jardinera usar los factores de 72 para decidir qué cuadrículas rectangulares usarán las 72 cápsulas de semillas?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Prime Factorization** — *Factorización prima*
- ☐ **Prime number** — *Número primo*
- ☐ **Composite number** — *Número compuesto*
- ☐ **Factor** — *Factor*
- ☐ **Factor tree** — *Árbol de factores*

Model: *60 is composite because it has more than two factors.* — Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Because 72 = ____, I can multiply combinations of those prime factors to get factors like ____, so a grid that works is ____ by ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Prime Factorization
2. **Notes 2:** Prime number
3. **Notes 3:** Composite number
4. **Notes 4:** Factor
5. **Notes 5:** Factor tree
6. **Notes 6:** Exponent
7. **Watch for this mistake:** *A common mistake in Prime Factorization is stopping the factor tree too soon — leaving a composite number like 4, 6, 8, or 9 in the final answer instead of continuing to split it. For example, writing $20 = 4 \times 5$ as the prime factorization (4 is composite) instead of finishing with $20 = 2 \times 2 \times 5$. Before you submit, check every number in your final answer: if any factor can still be broken into smaller factors, the tree isn't finished yet.*
8. **Try It 1:** A. 36 — *$2^2 \times 3^2 = 4 \times 9 = 36$, so 36 is the code for this panel.*
9. **Try It 2:** A. $2 \times 3 \times 13$ — *$78 = 2 \times 39 = 2 \times 3 \times 13$. All three factors are prime, so $2 \times 3 \times 13$ unlocks the final panel.*